

Assignment-2 Session 20

Question 1: Create a table comparing Agile and Waterfall models across 5 factors relevant to software testing, such as requirements, feedback cycle, tester involvement, documentation, and flexibility.

Answer

Factor	Agile Model	Waterfall Model
Requirements	Requirements can change throughout the project.	Requirements are fixed before development begins.
Feedback Cycle	Continuous feedback is received after every sprint.	Feedback is usually received after the complete project is developed.
Tester Involvement	Testers are involved from the beginning of each sprint and work closely with developers.	Testers usually start testing after the development phase is completed.
Documentation	Lightweight documentation with focus on working software.	Detailed documentation is prepared for every phase.
Flexibility	Highly flexible and easily adapts to changing customer requirements.	Less flexible; changes are difficult and costly after development starts.

Question 2: Pick any app you use daily (like Instagram, Zomato, or Paytm) and describe how the testing process would look different if the team used Agile instead of Waterfall.

Answer

Application: Zomato

If Zomato followed the **Agile Model**, testing would be performed continuously during every sprint instead of waiting until development is complete. Testers and developers would work together throughout the development process, and feedback would be incorporated quickly.

Changes in Testing Activities

Continuous Testing

Testers execute test cases during every sprint instead of waiting for the final build.

Sprint-wise Regression Testing

Regression testing is performed after every sprint to ensure existing features continue to work correctly.

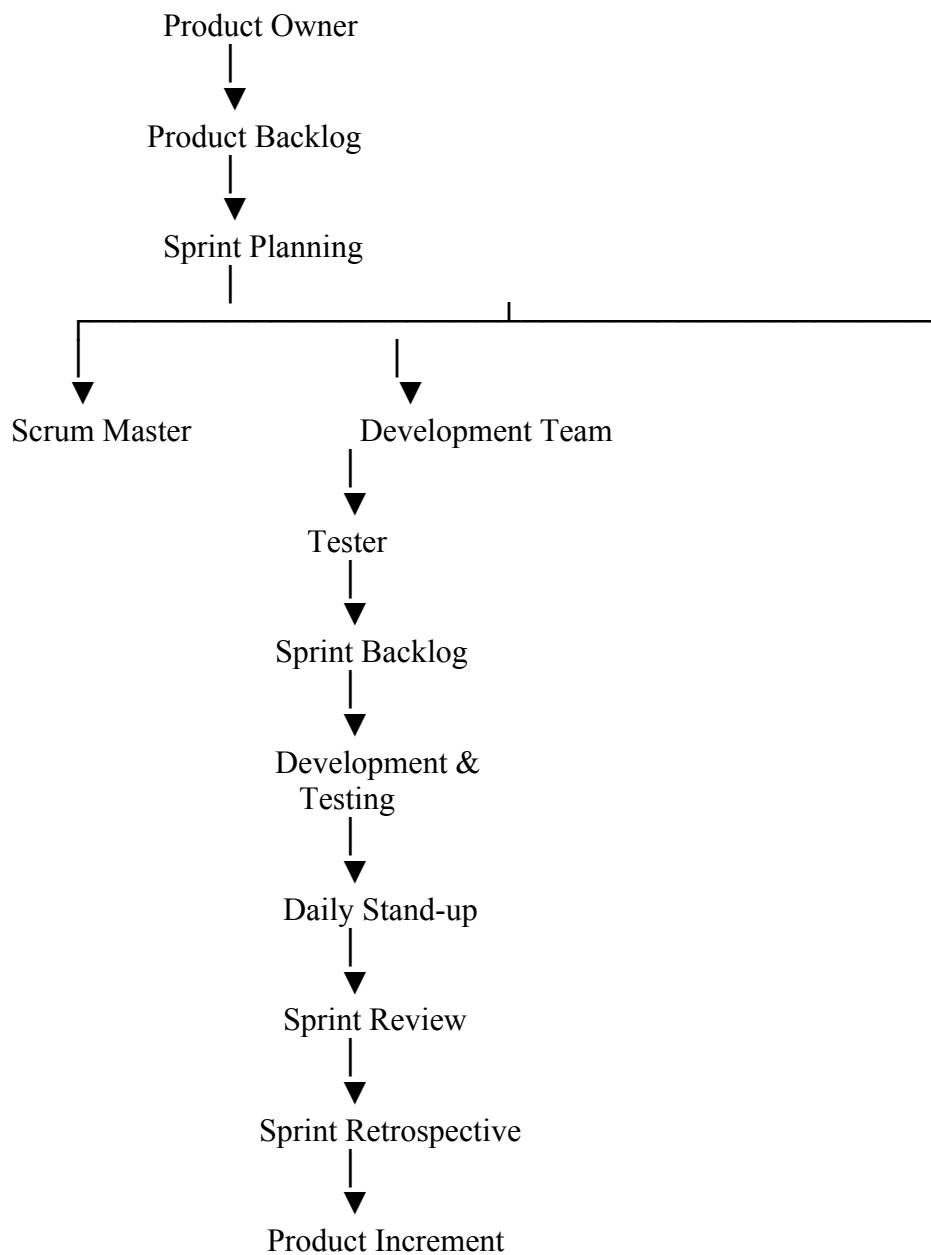
Daily Collaboration

Testers participate in Daily Scrum meetings with developers and the Product Owner to discuss progress, blockers, and defects.

Question 3: Draw a simple workflow diagram showing the main roles and ceremonies in a Scrum process.

Answer

Scrum Workflow Diagram



Scrum Roles

Product Owner: Defines requirements and prioritizes the Product Backlog.

Scrum Master: Facilitates Scrum activities and removes team blockers.

Development Team: Develops the software features.

Tester: Designs test cases, executes testing, reports defects, and performs regression testing.

Scrum Ceremonies

- Sprint Planning
- Daily Stand-up
- Sprint Review
- Sprint Retrospective

Question 4: For a food delivery app (like Swiggy), map out which testing activities should happen during each phase of a typical Agile sprint.

Answer

Application: Swiggy

Agile Sprint Phase	Testing Activities
Backlog Grooming	Review user stories, clarify acceptance criteria, identify test scenarios, and estimate testing effort.
Sprint Planning	Prepare the test plan, identify test data, estimate testing tasks, and assign responsibilities.
Development Phase	Review user stories, design test cases, prepare test data, and coordinate with developers.
During Development	Execute functional tests on completed stories, perform re-testing after bug fixes, and report defects.
Regression Testing	Verify that newly developed features have not impacted existing functionality.
Sprint Review	Demonstrate tested features to stakeholders, verify acceptance criteria, and collect feedback.
Sprint Retrospective	Discuss testing challenges, identify process improvements, and plan actions for the next sprint.

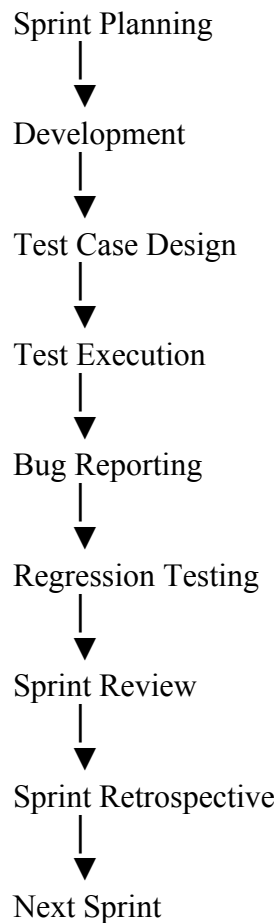
Agile Testing Life Cycle

Product Backlog



Backlog Grooming





Advantages of Agile Testing

- Continuous testing throughout development.
- Faster defect detection and resolution.
- Better collaboration between developers, testers, and business teams.
- Frequent customer feedback improves product quality.
- Faster software delivery through short sprint cycles.